

$$T: V \rightarrow W$$

* Operations of sets: Unions: $\cup$

Intersections: $\cap$

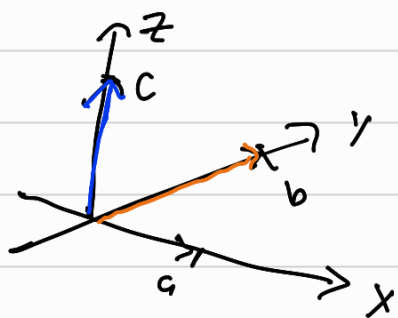
$$\bullet \text{span}\{e_1, \dots, e_n\}$$

$$\leadsto V = \mathbb{R}^3 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$$

$$\bullet S = \{(a, 0, 0), (0, b, 0), (0, 0, c)\} \quad \bullet \text{vectors}$$

$$\text{Span}(S) = n_1(a, 0, 0) + n_2(0, b, 0) + n_3(0, 0, c) \\ = (n_1 a, n_2 b, n_3 c) *$$

$$\bullet (1, 0, 0), (0, 1, 0), (0, 0, 1) \leftarrow \hat{e}_1, \hat{e}_2, \hat{e}_3$$



$$\text{Span}(S) = \mathbb{R}^3$$

* Notation: $\hat{e}_1 = (0, 0, 1)$ $\xrightarrow{\text{Linear Alg.}}$ $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \mid \text{Vectors hats: } \vec{x}$

operations: $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3 *$

$$T(a\vec{x} + b\vec{y}) = aT(\vec{x}) + bT(\vec{y})$$

$$\leadsto T(\vec{x}) = T\left(\sum_{i=1}^3 x_i \hat{e}_i\right) = T\left(x_1 \hat{e}_1 + x_2 \hat{e}_2 + x_3 \hat{e}_3\right)$$

$(1, 0, 0) \quad (0, 1, 0) \quad (0, 0, 1)$

$$\leadsto T\left(x_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right) = x_1 T\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + x_2 T\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + x_3 T\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$* T(\hat{e}_i) = T_i * \leftarrow \text{implies}$$

$$= \sum_{j=1}^3 T_{ij} \hat{u}_j$$

Basis have hats
 $\hat{e}, \hat{w}, \hat{u}, \dots$

why? $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ we get a vector

$T(\hat{e}_i)$ is a vector

$$T(\hat{e}_i) = \sum_{j=1}^3 T_{ij} \hat{u}_j \leftarrow \text{which I denote above } \hat{u}_j$$



$\vec{x}$: vector

$$T\left(\sum_{i=1}^3 x_i \hat{e}_i\right) = \sum_{i=1}^3 x_i T(\hat{e}_i) = \sum_{i=1}^3 x_i \left[\sum_{j=1}^3 T_{ij} \hat{u}_j \right]$$

$$= \sum_{i=1}^3 \sum_{j=1}^3 x_i T_{ij} \hat{u}_j = \sum_{j=1}^3 \left(\sum_{i=1}^3 x_i T_{ij} \right) \hat{u}_j$$

* In general: $\hat{u}$ was some "other" basis but we

can use $\hat{e}$ for $\hat{u}$.

$$\rightarrow \sum_{j=1}^3 \left(\sum_{i=1}^3 x_i T_{ij} \right) \hat{e}_j$$

T_{ij} : Are the Matrices

$i, j = 1, \dots, 3$

$$T = \begin{pmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{pmatrix}$$

• Now,

$\sum_{i=1}^3 x_i T_{ij}$: Matrix Multiplication

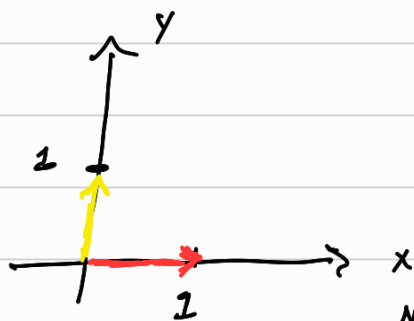
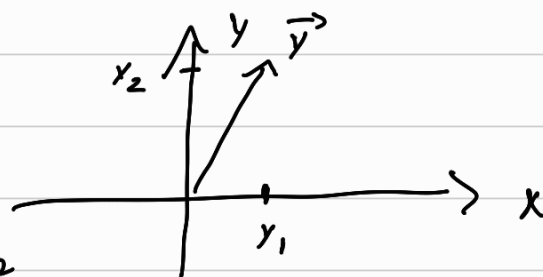
Aside: $\vec{y} \in \mathbb{R}^2$: $\vec{y} = (y_1, y_2)$

$$\vec{y} = (y_1, 0) + (0, y_2)$$

$$= y_1 (1, 0) + y_2 (0, 1) = y_1 \hat{e}_1 + y_2 \hat{e}_2$$

$$= \sum_{i=1}^2 y_i \hat{e}_i \quad *$$

Basis Vectors of $\mathbb{R}^2$



notation

$$* \boxed{T(\hat{e}_i)} \equiv T_i$$

T_i is a vector in $\mathbb{R}^3$

$T(\vec{x}) \leftarrow$ Belongs to $\mathbb{R}^3$

$$\left(\underline{f(x)} = \underline{x^2 - 2} \right)$$

$$\rightarrow T(\vec{x}) = T\left(\sum_{i=1}^3 x_i \hat{e}_i\right) = \sum_{i=1}^3 x_i T(\hat{e}_i)$$

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(\hat{e}_i) = T_i$$

Scalar

3

$$\vec{x} = \sum_{i=1}^3 x_i \hat{e}_i$$

vector

